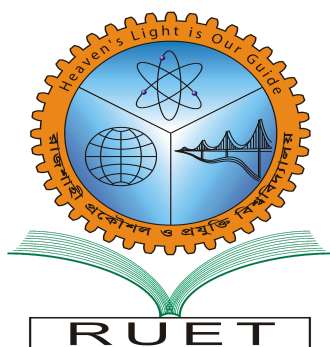


"Heaven's light is our guide"



Department of Computer Science & Engineering

Rajshahi University of Engineering & Technology

LAB REPORT

Course Code : CSE 3102.

Course Name : Database Sessional.

Report No. : 2

Report Name : **SQL codes** for lab final selections.

By: Name : Md. Rraf Hasan Roll : 2103062 Section : B Series : 21	To: Md. Azmain Yakin Srizon Assistant Professor, Department of Computer Science & Engineering , RUET.
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Submission date : 18 Jun, 2025.

Database: Versity Management .

Schemas:

```
CREATE TABLE Students (  
    student_id INT AUTO_INCREMENT PRIMARY KEY,  
    student_name VARCHAR(100) NOT NULL,  
    email VARCHAR(100),  
    phone VARCHAR(20),  
    address VARCHAR(100),  
    date_of_birth DATE,  
    gender ENUM('Male', 'Female', 'Other'),  
    enrollment_date DATE,  
    guardian_name VARCHAR(100),  
    guardian_phone_no VARCHAR(20)  
);  
  
CREATE TABLE Teachers (  
    teacher_id INT AUTO_INCREMENT PRIMARY KEY,  
    teacher_name VARCHAR(100) NOT NULL,  
    email VARCHAR(100),  
    phone VARCHAR(20),  
    address VARCHAR(200),  
    date_of_birth DATE,  
    gender ENUM('Male', 'Female', 'Other'),  
    joining_date DATE,  
    qualification VARCHAR(100),  
    specialization VARCHAR(100)  
);  
  
CREATE TABLE Courses (  
    course_id INT AUTO_INCREMENT PRIMARY KEY,  
    course_name VARCHAR(100) NOT NULL,  
    course_code VARCHAR(20) UNIQUE,  
    course_credit DECIMAL(3,1),  
    description TEXT,  
    teacher_id INT,  
    FOREIGN KEY (teacher_id) REFERENCES Teachers(teacher_id)  
);
```

```
CREATE TABLE Departments (
    department_id INT AUTO_INCREMENT PRIMARY KEY,
    department_name VARCHAR(100) NOT NULL,
    department_head VARCHAR(100),
    teacher_id INT,
    FOREIGN KEY (teacher_id) REFERENCES Teachers(teacher_id)
);
```

```
CREATE TABLE Enrollments (
    enrollment_id INT AUTO_INCREMENT PRIMARY KEY,
    student_id INT,
    course_id INT,
    enrollment_date DATE,
    FOREIGN KEY (student_id) REFERENCES Students(student_id),
    FOREIGN KEY (course_id) REFERENCES Courses(course_id)
);
```

Students:

-- Inserting sample data into Students table

```
INSERT INTO
Students (student_name, email, phone, address, date_of_birth, gender,
enrollment_date, guardian_name, guardian_phone_no)
VALUES
('John Smith', 'john.smith@example.com', '555-123-4567', '123 Main St, Anytown',
'2005-03-15', 'Male', '2020-09-01', 'Robert Smith', '555-987-6543'),
('Emily Johnson', 'emily.j@example.com', '555-234-5678', '456 Oak Ave, Somewhere',
'2004-07-22', 'Female', '2019-09-01', 'Susan Johnson', '555-876-5432'),
('Michael Brown', 'michael.b@example.com', '555-345-6789', '789 Pine Rd, Nowhere',
'2006-01-30', 'Male', '2021-09-01', 'David Brown', '555-765-4321'),
('Sarah Williams', 'sarah.w@example.com', '555-456-7890', '321 Elm St, Anywhere',
'2005-11-12', 'Female', '2020-09-01', 'Jennifer Williams', '555-654-3210'),
('James Davis', 'james.d@example.com', '555-567-8901', '654 Maple Dr, Everywhere',
'2004-09-05', 'Male', '2019-09-01', 'Thomas Davis', '555-543-2109'),
('Jessica Miller', 'jessica.m@example.com', '555-678-9012', '987 Cedar Ln,
Somewhere', '2006-02-18', 'Female', '2021-09-01', 'Patricia Miller',
'555-432-1098'),
('Robert Wilson', 'robert.w@example.com', '555-789-0123', '135 Birch Blvd,
Nowhere', '2005-06-25', 'Male', '2020-09-01', 'Charles Wilson', '555-321-0987'),
('Jennifer Taylor', 'jennifer.t@example.com', '555-890-1234', '246 Walnut Way,
Anywhere', '2004-12-08', 'Female', '2019-09-01', 'Karen Taylor', '555-210-9876'),
('Daniel Anderson', 'daniel.a@example.com', '555-901-2345', '369 Spruce Ct,
Everywhere', '2006-04-03', 'Male', '2021-09-01', 'Paul Anderson', '555-109-8765'),
('Lisa Thomas', 'lisa.t@example.com', '555-012-3456', '482 Ash Pl, Somewhere',
'2005-08-19', 'Female', '2020-09-01', 'Nancy Thomas', '555-098-7654'),
```

```
('Matthew Jackson', 'matthew.j@example.com', '555-123-4567', '591 Poplar Dr,
Nowhere', '2004-10-27', 'Male', '2019-09-01', 'Mark Jackson', '555-987-6543'),
('Amanda White', 'amanda.w@example.com', '555-234-5678', '753 Willow Ln,
Anywhere', '2006-05-14', 'Female', '2021-09-01', 'Laura White', '555-876-5432'),
('Christopher Harris', 'chris.h@example.com', '555-345-6789', '864 Sycamore Ave,
Everywhere', '2005-07-21', 'Male', '2020-09-01', 'Steven Harris', '555-765-4321'),
('Nicole Martin', 'nicole.m@example.com', '555-456-7890', '975 Palm St,
Somewhere', '2004-11-30', 'Female', '2019-09-01', 'Betty Martin', '555-654-3210'),
('Andrew Thompson', 'andrew.t@example.com', '555-567-8901', '1086 Olive Rd,
Nowhere', '2006-03-09', 'Male', '2021-09-01', 'Richard Thompson', '555-543-2109');
```

Teachers:

-- Inserting sample data into Teachers table

INSERT INTO Teachers

(teacher_name, email, phone, address, date_of_birth, gender, joining_date, qualification, specialization)

VALUES

```
('Dr. Susan Parker', 's.parker@example.com', '555-111-2222', '111 University Dr,
Anytown', '1975-04-12', 'Female', '2010-08-15', 'PhD in Mathematics', 'Advanced
Calculus'),
('Prof. Mark Johnson', 'm.johnson@example.com', '555-222-3333', '222 College Ave,
Somewhere', '1980-09-23', 'Male', '2012-03-01', 'MSc in Physics', 'Quantum
Mechanics'),
('Dr. Lisa Chen', 'l.chen@example.com', '555-333-4444', '333 Campus Ln, Nowhere',
'1978-11-05', 'Female', '2015-01-10', 'PhD in Chemistry', 'Organic Chemistry'),
('Prof. David Wilson', 'd.wilson@example.com', '555-444-5555', '444 School St,
Anywhere', '1982-06-18', 'Male', '2018-09-05', 'MA in English', 'American
Literature'),
('Dr. Rachel Lee', 'r.lee@example.com', '555-555-6666', '555 Academy Blvd,
Everywhere', '1973-12-30', 'Female', '2008-07-20', 'PhD in Biology', 'Genetics'),
('Prof. Brian Adams', 'b.adams@example.com', '555-666-7777', '666 Education Dr,
Somewhere', '1985-02-14', 'Male', '2019-08-25', 'MSc in Computer Science',
'Artificial Intelligence'),
('Dr. Karen Brown', 'k.brown@example.com', '555-777-8888', '777 Learning Way,
Nowhere', '1976-08-09', 'Female', '2011-05-15', 'PhD in History', 'European
History'),
('Prof. Steven Taylor', 's.taylor@example.com', '555-888-9999', '888 Knowledge Ct,
Anywhere', '1983-03-27', 'Male', '2017-02-10', 'MA in Economics',
'Macroeconomics'),
('Dr. Patricia Clark', 'p.clark@example.com', '555-999-0000', '999 Wisdom Pl,
Everywhere', '1970-07-19', 'Female', '2005-09-01', 'PhD in Psychology', 'Cognitive
Psychology'),
('Prof. Kevin Martinez', 'k.martinez@example.com', '555-000-1111', '1010 Insight
Dr, Somewhere', '1984-10-31', 'Male', '2016-11-15', 'MSc in Engineering',
'Mechanical Engineering'),
```

```
( 'Dr. Nancy Scott', 'n.scott@example.com', '555-111-0000', '1212 Discovery Ln,
Nowhere', '1979-05-22', 'Female', '2014-04-05', 'PhD in Political Science',
'International Relations'),
( 'Prof. Jason Green', 'j.green@example.com', '555-222-1111', '1313 Innovation Ave,
Anywhere', '1981-01-15', 'Male', '2013-08-20', 'MA in Fine Arts', 'Painting and
Sculpture'),
( 'Dr. Michelle Hall', 'm.hall@example.com', '555-333-2222', '1414 Creativity Rd,
Everywhere', '1974-09-08', 'Female', '2009-10-10', 'PhD in Music', 'Music
Theory'),
( 'Prof. Eric Young', 'e.young@example.com', '555-444-3333', '1515 Imagination
Blvd, Somewhere', '1986-12-03', 'Male', '2020-01-15', 'MSc in Environmental
Science', 'Climate Change'),
( 'Dr. Angela King', 'a.king@example.com', '555-555-4444', '1616 Inspiration Way,
Nowhere', '1977-04-17', 'Female', '2012-07-01', 'PhD in Philosophy', 'Ethics');
```

Courses:

-- Inserting sample data into Courses table

```
INSERT INTO
Courses (course_name, course_code, course_credit, description, teacher_id)
VALUES
( 'Calculus I', 'MATH101', 4.0, 'Introduction to differential and integral
calculus', 1),
( 'Physics I', 'PHYS101', 4.0, 'Fundamentals of mechanics and thermodynamics', 2),
( 'Organic Chemistry', 'CHEM201', 3.0, 'Structure, properties, and reactions of
organic compounds', 3),
( 'American Literature', 'ENGL202', 3.0, 'Survey of American literature from
colonial times to present', 4),
( 'Genetics', 'BIOL301', 4.0, 'Principles of heredity and variation in organisms',
5),
( 'Introduction to AI', 'CS401', 3.0, 'Fundamental concepts of artificial
intelligence', 6),
( 'European History', 'HIST102', 3.0, 'Survey of European history from Renaissance
to present', 7),
( 'Macroeconomics', 'ECON201', 3.0, 'Study of economy-wide phenomena', 8),
( 'Cognitive Psychology', 'PSYC302', 3.0, 'Study of mental processes such as
attention and memory', 9),
( 'Mechanical Engineering', 'ENGR101', 4.0, 'Fundamentals of mechanical engineering
principles', 10),
( 'International Relations', 'POLS203', 3.0, 'Study of relationships between
countries', 11),
( 'Painting Techniques', 'ART105', 2.0, 'Introduction to various painting methods
and styles', 12),
( 'Music Theory', 'MUS101', 3.0, 'Fundamentals of music notation and composition',
13),
```

```
( 'Environmental Science', 'ENVS201', 3.0, 'Interdisciplinary study of
environmental systems', 14),
( 'Ethics', 'PHIL101', 3.0, 'Study of moral philosophy and decision making', 15),
( 'Calculus II', 'MATH102', 4.0, 'Continuation of Calculus I with advanced topics',
1),
( 'Physics II', 'PHYS102', 4.0, 'Electricity, magnetism, and optics', 2);
```

Departments:

-- Inserting sample data into Departments table

```
INSERT INTO
Departments (department_name, department_head, teacher_id)
VALUES
( 'Mathematics', 'Dr. Susan Parker', 1),
( 'Physics', 'Prof. Mark Johnson', 2),
( 'Chemistry', 'Dr. Lisa Chen', 3),
( 'English', 'Prof. David Wilson', 4),
( 'Biology', 'Dr. Rachel Lee', 5),
( 'Computer Science', 'Prof. Brian Adams', 6),
( 'History', 'Dr. Karen Brown', 7),
( 'Economics', 'Prof. Steven Taylor', 8),
( 'Psychology', 'Dr. Patricia Clark', 9),
( 'Engineering', 'Prof. Kevin Martinez', 10),
( 'Political Science', 'Dr. Nancy Scott', 11),
( 'Fine Arts', 'Prof. Jason Green', 12),
( 'Music', 'Dr. Michelle Hall', 13),
( 'Environmental Studies', 'Prof. Eric Young', 14),
( 'Philosophy', 'Dr. Angela King', 15);
```

Enrollments:

```
INSERT INTO
Enrollments (enrollment_id, student_id, course_id, enrollment_date)
VALUES
(1, 1, 1, '2023-09-01'),
(2, 2, 1, '2023-09-01'),
(3, 3, 1, '2023-09-01'),
(4, 1, 16, '2023-09-01'),
(5, 4, 16, '2023-09-01'),
(6, 5, 2, '2023-09-01'),
(7, 6, 3, '2023-09-01'),
(8, 7, 4, '2023-09-01'),
(9, 8, 5, '2023-09-01'),
(10, 9, 6, '2023-09-01'),
(11, 10, 7, '2023-09-01'),
```

```
(12, 11, 8, '2023-09-01'),
(13, 12, 9, '2023-09-01'),
(14, 13, 10, '2023-09-01'),
(15, 14, 11, '2023-09-01'),
(16, 15, 12, '2023-09-01'),
(17, 1, 2, '2023-09-01'),
(18, 2, 3, '2023-09-01'),
(19, 3, 4, '2023-09-01'),
(20, 4, 5, '2023-09-01');
```

Queries:

-> Find all students enrolled in courses taught by Dr. Susan Parker

sol: SELECT s.student_name, c.course_name
FROM Students s
JOIN Enrollments e ON s.student_id = e.student_id
JOIN Courses c ON e.course_id = c.course_id
JOIN Teachers t ON c.teacher_id = t.teacher_id
WHERE t.teacher_name = 'Dr. Susan Parker';

student_name	course_name
John Smith	Calculus I
John Smith	Calculus II
Emily Johnson	Calculus I
Michael Brown	Calculus I
Sarah Williams	Calculus II

-> Count the no. of courses in each department

sol: SELECT d.department_name, COUNT(c.course_id) AS course_count
FROM Departments d
LEFT JOIN Teachers t ON d.teacher_id = t.teacher_id
LEFT JOIN Courses c ON t.teacher_id = c.teacher_id
GROUP BY d.department_name
ORDER BY course_count;

department_name	course_count ▲ 1
Engineering	1
History	1
Philosophy	1
English	1
Fine Arts	1
Psychology	1
Computer Science	1
Environmental Studies	1
Chemistry	1
Political Science	1
Economics	1
Biology	1
Music	1
Physics	2
Mathematics	2

-> Find teachers who teach more than 1 course

sol: SELECT t.teacher_name, COUNT(c.course_id) AS courses_taught
FROM Teachers t
JOIN Courses c ON t.teacher_id = c.teacher_id
GROUP BY t.teacher_id
HAVING COUNT(c.course_id) > 1
ORDER BY courses_taught;

teacher_name	courses_taught ▲ 1
Prof. Mark Johnson	2
Dr. Susan Parker	2

-> List courses with more than 3 credits along with teacher information

sol: SELECT c.course_name, c.course_credit, t.teacher_name, t.specialization
FROM Courses c
JOIN Teachers t ON c.teacher_id = t.teacher_id
WHERE c.course_credit > 3
ORDER BY c.course_credit;

course_name	course_credit ▲ 1	teacher_name	specialization
Calculus I	4.0	Dr. Susan Parker	Advanced Calculus
Physics I	4.0	Prof. Mark Johnson	Quantum Mechanics
Genetics	4.0	Dr. Rachel Lee	Genetics
Mechanical Engineering	4.0	Prof. Kevin Martinez	Mechanical Engineering
Calculus II	4.0	Dr. Susan Parker	Advanced Calculus
Physics II	4.0	Prof. Mark Johnson	Quantum Mechanics

-> Find the most popular course

sol: SELECT c.course_name, COUNT(e.student_id) AS enrollment_count
FROM Courses c
JOIN Enrollments e ON c.course_id = e.course_id
GROUP BY c.course_id
ORDER BY enrollment_count DESC
LIMIT 1;

course_name	enrollment_count
Calculus I	3

-> List departments with the average years of experience of their teachers

sol: SELECT d.department_name,
AVG(TIMESTAMPDIFF(YEAR, t.joining_date, CURDATE())) AS avg_experience_years
FROM Departments d
JOIN Teachers t ON d.teacher_id = t.teacher_id
GROUP BY d.department_name
ORDER BY avg_experience_years DESC;

department_name	avg_experience_years ▼ 1
Psychology	19.0000
Biology	16.0000
Music	15.0000
Mathematics	14.0000
History	14.0000
Physics	13.0000
Philosophy	12.0000
Fine Arts	11.0000
Political Science	11.0000
Chemistry	10.0000
Economics	8.0000
Engineering	8.0000
English	6.0000
Environmental Studies	5.0000
Computer Science	5.0000

-> List the teachers with PhD holder

sol: SELECT
t.teacher_id,
t.teacher_name,
t.email,
t.phone,
t.qualification,
t.specialization,
d.department_name,
t.joining_date,
TIMESTAMPDIFF(YEAR, t.joining_date, CURDATE()) AS years_of_experience
FROM Teachers t
LEFT JOIN Departments d ON t.teacher_id = d.teacher_id
WHERE t.qualification REGEXP 'PhD|Ph\\.D|Doctorate'
ORDER BY t.teacher_name;

teacher_id	teacher_name ▲ 1	email	phone	qualification	specialization	department_name	joining_date	years_of_experience
15	Dr. Angela King	a.king@example.com	555-555-4444	PhD in Philosophy	Ethics	Philosophy	2012-07-01	12
7	Dr. Karen Brown	k.brown@example.com	555-777-8888	PhD in History	European History	History	2011-05-15	14
3	Dr. Lisa Chen	l.chen@example.com	555-333-4444	PhD in Chemistry	Organic Chemistry	Chemistry	2015-01-10	10
13	Dr. Michelle Hall	m.hall@example.com	555-333-2222	PhD in Music	Music Theory	Music	2009-10-10	15
11	Dr. Nancy Scott	n.scott@example.com	555-111-0000	PhD in Political Science	International Relations	Political Science	2014-04-05	11
9	Dr. Patricia Clark	p.clark@example.com	555-999-0000	PhD in Psychology	Cognitive Psychology	Psychology	2005-09-01	19
5	Dr. Rachel Lee	r.lee@example.com	555-555-6666	PhD in Biology	Genetics	Biology	2008-07-20	16
1	Dr. Susan Parker	s.parker@example.com	555-111-2222	PhD in Mathematics	Advanced Calculus	Mathematics	2010-08-15	14

-> List all students born after 2005

sol: SELECT s.student_name, s.date_of_birth, s.guardian_name, s.guardian_phone_no
FROM Students s
WHERE s.date_of_birth > '2005-12-31'
ORDER BY s.date_of_birth;

← T →	▼	student_name	date_of_birth ▲ 1	guardian_name	guardian_phone_no
☐	Edit Copy Delete	Michael Brown	2006-01-30	David Brown	555-765-4321
☐	Edit Copy Delete	Jessica Miller	2006-02-18	Patricia Miller	555-432-1098
☐	Edit Copy Delete	Andrew Thompson	2006-03-09	Richard Thompson	555-543-2109
☐	Edit Copy Delete	Daniel Anderson	2006-04-03	Paul Anderson	555-109-8765
☐	Edit Copy Delete	Amanda White	2006-05-14	Laura White	555-876-5432